

Fair, Non-Progressive Influence Control Under Backfire and Parameter Uncertainty

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Abstract—The classical influence maximization guarantee of Kempe, Kleinberg, and Tardos rests on four assumptions: progressive diffusion, a monotone submodular objective, known influence parameters, and no fairness requirement. Real deployments routinely violate all four at once: influence decays and can be actively reversed (backfire, or negative word-of-mouth), edge probabilities are never known exactly, and social interventions must not neglect a disadvantaged subpopulation. This paper studies the combined relaxation. We prove that a sandwich approximation built from two monotone-submodular surrogates yields a $\rho(1 - 1/e)$ guarantee for *every* backfire intensity $\beta \in [0, 1]$, degenerating gracefully to the tight $1 - 1/e$ bound at $\beta = 0$ and admitting a per-instance, directly computable certificate. We derive a closed-form Lagrangian index for the resulting 2-state/3-action per-node control problem, replacing a bisection-plus-value-iteration inner loop with an $O(Km + n \log n)$ -per-round procedure that we measure to be roughly two orders of magnitude faster than the bisection oracle it replaces. We show that $(1 - 1/e)$ -hardness survives backfire exactly. Finally, we identify and correct a population-weighting pathology in an existing isoelastic-welfare fairness objective that systematically under-protects small groups, and replace it with an egalitarian reweighting. We implement all of this in a control policy, MF-BWI-Fair, that combines mean-field belief propagation, Bayesian parameter tracking, the closed-form index, and the corrected fairness reweighting. Experiments on a 120-node synthetic stochastic block model show MF-BWI-Fair sustaining spread under backfire and recovery where one-shot baselines – including independently reimplemented published algorithms (IMM, He–Kempe’s robust Saturate Greedy) – degrade sharply, reaching over $6\times$ the best one-shot baseline’s spread at the most severe recovery rate tested, and show the egalitarian fix raising worst-off-group reach at negligible spread cost. We validate generalization on three further synthetic graphs (two scales, a structurally different scale-free topology) and one real social network (a 224-node Facebook ego-network): the synthetic-graph finding reproduces cleanly, while the real network initially showed no clear advantage under a necessarily reduced compute budget; restoring the main benchmark’s compute budget on the identical real graph flips every tested checkpoint from behind to tied-or-ahead, indicating the earlier divergence was substantially a budget artifact rather than a structural failure, albeit with a narrower margin than on synthetic graphs. We additionally benchmark our IMM reimplementation against the original authors’ own C++ code and find it trustworthy (within 0.6% expected spread).

Index Terms—Influence maximization, submodular optimization, restless bandits, fairness, robust optimization, social networks.

I. INTRODUCTION

Kempe, Kleinberg, and Tardos [1] showed that under the Independent Cascade (IC) and Linear Threshold (LT) models, the expected spread of a seed set is monotone and submodular, so the greedy algorithm achieves a $(1 - 1/e)$ -approximation via the general result of Nemhauser, Wolsey, and Fisher [2], and this ratio is tight under standard complexity assumptions [3]. That guarantee, and the large body of scalable algorithms it spawned – CELF’s lazy-forward speedup [4] and the near-linear-time RIS/martingale algorithm IMM [5], [6] chief among them – rests on four assumptions taken together: (A1) diffusion is *progressive* (once active, always active); (A2) the spread function is *monotone submodular*; (A3) influence parameters are *known* exactly; and (A4) the objective has *no fairness structure* – every node’s activation counts equally toward one aggregate number.

Each assumption is routinely false in the settings influence maximization is used to model. Interventions decay: a vaccinated or informed individual can relapse or forget, and messaging campaigns require sustained reinforcement, not a one-time seeding. Influence can be actively *negative*: word of mouth can backfire, and an over-exposed contact can turn an active neighbor against the message rather than merely fail to convert it [7], [8], [9]. Edge probabilities are estimated from noisy or incomplete cascade data, never known exactly. And in the social-intervention settings that motivate much of this work in the first place – HIV-prevention and stable-housing information campaigns among homeless youth, the application context Rahmattalabi et al. build their fairness-aware formulation around [10] – treating every node as interchangeable is not a simplification but a policy failure: a mechanism that maximizes aggregate reach while systematically under-serving one subpopulation is not simply “unfair” in the abstract, it directly undermines the deployment’s stated public-health goal.

Prior work has relaxed these assumptions one, or occasionally two, at a time (Section II). This paper studies all four relaxations *together*: a model with recovery and backfire-driven deactivation (relaxing A1 and, as we show, A2), Bayesian tracking of unknown influence and recovery parameters (relaxing A3), and an egalitarian-welfare fairness objective over groups (relaxing A4). Our literature search found no existing paper that combines all four axes; the two closest works each combine exactly two (Section II).

Contributions

- **A formal layered-semantics model** (Section III) in which an active node survives a round unless it spontaneously recovers or is backfired by an active in-neighbor, and any node – active or not – can be (re-)activated by a successful positive trial from an active in-neighbor.
- **A proven submodularity result for recovery alone** (Lemma 2): spontaneous, state-independent recovery preserves monotone submodularity of cumulative spread on any graph, via a layered live-edge-graph reachability argument. This isolates recovery as the “free” relaxation; neighbor-dependent backfire is what destroys the coverage structure.
- **A sandwich approximation theorem** (Theorem 1, *proven*, instantiating the general sandwich technique of [11] together with [2]): for *every* backfire intensity $\beta \in [0, 1]$, a greedy solution taken over two monotone-submodular surrogates and the raw objective achieves $\rho(1 - 1/e) \text{OPT}_\beta$, where ρ is an instance-dependent sandwich ratio that degenerates to the tight $1 - 1/e$ bound at $\beta = 0$. The guarantee is also directly instance-certifiable: a returned set ships its own approximation ratio, computable by simulating the two $\beta = 0$ surrogates.
- **A closed-form Lagrangian index** (Propositions 2–4, Theorem 2, *proven* except for indexability itself, which is checked numerically and explicitly flagged as such) for the resulting 2-state/3-action per-node control problem, reducing the per-round decision from bisection with per-node value iteration to $O(Km + n \log n)$ – an approximately $130\times$ speedup measured in this project’s implementation.
- **A hardness result that preserves honest status labels**: Theorem 3 *proves* that $(1 - 1/e)$ -inapproximability survives backfire exactly, for every fixed β . A further, β -dependent constant-factor worsening is numerically plausible but remains an open conjecture, not a claimed result of this paper (Section VIII).
- **A corrected fairness mechanism**: we identify a measured population-weighting pathology in the isoelastic welfare objective of [10] (small groups are protected only once under-served by a factor proportional to a group-size ratio) and replace it with an egalitarian (population-unweighted) reweighting that protects the worst-off group at a neutral default setting.
- **MF-BWI-Fair**, a control policy combining mean-field belief propagation, Bayesian posterior tracking, the closed-form Lagrangian index, and the corrected welfare reweighting, together with a preliminary comparison against six algorithms including independently-verified reimplementations of two published baselines.

II. RELATED WORK

A. Classical influence maximization

Kempe, Kleinberg, and Tardos [1] established monotone submodularity of expected spread under IC/LT and the resulting $(1 - 1/e)$ greedy guarantee [2], tight under P $\neq$

NP [3]. CELF [4] gives a lazy-forward speedup exploiting the same submodularity. IMM [5], building on the RIS/reverse-reachable-set duality of [6], gives an $O((k+\ell)(n+m) \log n/\varepsilon^2)$ near-linear-time $(1 - 1/e - \varepsilon)$ -approximation and is, to our knowledge, the most-cited scalable classical algorithm with publicly available original code [12].

B. Fair / group-fairness-constrained influence maximization

Tsang et al. [13] introduce maximin and diversity-constraint fairness notions as a multi-objective submodular problem. Fish et al. [14] study a maximin welfare objective directly and prove it is not monotone submodular in general. Farnadi et al. [15] unify several fairness notions as exact MILP formulations at small scale, with no submodular approximation guarantee. Becker et al. [16] show deterministic maximin-fair seeding is hard to approximate but that randomized seeding recovers a $(1 - 1/e - \varepsilon)$ -type guarantee. Rahmat-talabi et al. [10] propose an isoelastic (CES) welfare objective over per-group reach and prove monotone submodularity via the concave-nondecreasing-composed-with-monotone-submodular lemma of [17], giving an exact $(1 - 1/e)$ guarantee – but only for the classical progressive IC model; their proof does not address deactivation or negative influence. This paper adopts their objective *form*, not their guarantee, and additionally corrects a population-weighting side effect of their exact formula (Section V-A). Rui et al. [18] and Feng et al. [19] extend fairness to scalable and data-driven/learning-based settings, respectively; the latter is out of scope for the classical-algorithms focus here.

C. Non-progressive / recovery-capable influence maximization

Chan and Ning [20] show that non-progressive dynamics under the LT model lose submodularity on general graphs, recovering it only on DAGs. Golnari et al. [21] propose the “Heat Conduction” model, proving submodularity under a specially engineered non-progressive rule. Lou et al. [22] give a continuous-time two-state Markov estimation model without a submodularity result. Zahoor et al. [23] and Hui et al. [24] study reactivation dynamics, the latter via deep reinforcement learning (out of scope here). Taninmis et al. [25] study an adversarial deactivation game, a different framing from the spontaneous, non-adversarial backfire studied here. Our Lemma 2 gives a stronger/different result than Chan and Ning’s DAG-restricted finding: spontaneous, state-independent recovery alone preserves monotone submodularity on *any* graph, via layered-DAG reachability rather than threshold-rule structure.

D. Non-monotone / non-submodular / negative-influence models

Bharathi et al. [26] give the first competitive cascade model. Chen et al. [7] extend IC with a quality factor allowing activated nodes to flip negative, engineered to remain submodular. He et al. [8] introduce a two-competing-cascade “sandwich approximation” as a scalability device, whose general form is due to Lu, Chen, and Lakshmanan [11]; our Theorem 1

instantiates the latter’s general monotone-submodular bracket $L \leq f \leq U$, applied to a materially different bracket – a recovery-only process, not a two-cascade one. Li et al. [9] give a signed-network model (IC-P) that also remains submodular by construction. Buchbinder et al. [27] and Tukan et al. [28] give tight and practical constant-factor guarantees for unconstrained/cardinality-constrained *non-monotone submodular* maximization – a different relaxation axis (non-monotonicity of a still-submodular function) from ours. Bian et al. [29] give a submodularity-ratio/curvature-based greedy guarantee for non-submodular functions, superseded here by the sandwich approach, which holds for all β rather than requiring an estimated ratio. Horel and Singer [30] give an exponential query-complexity lower bound for functions ε -far from submodular once $\varepsilon = \omega(n^{-1/2})$, but only for an *adversarial* perturbation; our backfire perturbation is structured (a bounded extra kill rate), which is why Theorem 1 escapes this limitation (Section IV-D).

E. Bayesian, robust, and online influence maximization under uncertainty

He and Kempe [31] formalize worst-case-across-scenarios robust influence maximization and give a bicriteria Saturate Greedy algorithm with a provable budget-blowup/quality trade-off; no official code is available for this specific paper. The dominant paradigm for online uncertainty is combinatorial-bandit learning [32], [33], [34], [35], [36], [37] or one-shot distributionally robust optimization [38], [39]. A genuinely single-shot Bayesian formulation – integrating expected spread over a prior without adaptivity or worst-case hedging – exists only embedded inside bandit posterior-update machinery, not as a standalone anchor result; this is the gap our Bayesian tracker sits in.

F. Restless-bandit sequential control

Mate et al. [40] give binary-action Whittle-index results for collapsing bandits, explicitly leaving multi-action indexability future work. Killian, Perrault, and Tambe [41] give the Lagrangian-relaxation template for general multi-action restless bandits that our closed-form index specializes and simplifies for the specific 2-state/3-action case, avoiding the general indexability problem they call “notoriously difficult.” Ou et al. [42] propose an alternative to mean-field network coupling that we verified does not apply here: their coupling is through a deterministic, known action-vector, not through another arm’s actual random state, which is the genuinely bilinear coupling our setting requires. Li and Varakantham [43] study fairness constraints in independent-arm restless bandits, confirming fairness-in-RMAB as an active but separate line of work.

G. Positioning

No paper we found combines group fairness, non-progressive/recovery dynamics, non-monotone backfire influence, and Bayesian parameter uncertainty. The two closest works each combine exactly two of these four axes, both

very recent: a 2026 PACMMOD paper combining robust fairness with multiple-community-partition (parameter) uncertainty [44], and Fang et al.’s 2026 scalable fair influence *blocking* maximization, combining fairness with non-monotone/blocking-style negative influence [45]. Separately, no paper we found connects submodular/RIS-style influence-maximization theory to restless-bandit sequential control theory at all; Section IV-C is, to the depth of our search, the first to do so for a node-level activation/maintenance control problem.

III. PROBLEM FORMULATION

Let $G = (V, E)$ be a directed graph, $n = |V|$, $m = |E|$, with a partition of V into groups $G_1, \dots, G_M$. Rounds are $t = 0, 1, \dots, T$; the active set $A_t \subseteq V$ starts at $A_0 = S$, the seed set. Each round draws independent randomness: for every edge (u, v) , a positive trial $X_{uv}^t \sim \text{Bern}(p_{uv})$ and a backfire trial $Y_{uv}^t \sim \text{Bern}(\beta p_{uv})$, for a global backfire intensity $\beta \in [0, 1]$; for every node v , a recovery trial $R_v^t \sim \text{Bern}(q_v)$.

Transition (layered semantics). $v \in A_{t+1}$ iff

$$\begin{aligned} &[v \in A_t \wedge R_v^t = 0 \wedge \forall u \in A_t \cap N_{\text{in}}(v) : Y_{uv}^t = 0] \\ &\vee [\exists u \in A_t \cap N_{\text{in}}(v) : X_{uv}^t = 1]. \end{aligned} \quad (1)$$

An active node stays active unless it recovers or is backfired by an active in-neighbor, and any node – active or not – is (re-)activated by a successful positive trial from an active in-neighbor; in particular a positive trial can *rescue* an active node in the same round it would otherwise have recovered or been backfired. This rescue clause is exactly what makes the one-step map $A_t \mapsto A_{t+1}$ monotone in A_t at $\beta = 0$, which the submodularity result of Section IV rests on; for $\beta > 0$ the map is not monotone, since the backfire clause is anti-monotone.

The objective is cumulative expected spread,

$$f_\beta(S) = \mathbb{E} \left[\sum_{t=1}^T |A_t| \right], \quad A_0 = S, \quad (2)$$

time-averaged spread being the same quantity up to the constant $1/T$.

Fairness. Let u_g denote group g ’s time-averaged realized reach fraction (active fraction of G_g , averaged over rounds). Following an isoelastic (constant elasticity of substitution) welfare form [10], define

$$W_\alpha(u) = \sum_g N_g \frac{u_g^\alpha}{\alpha} \quad (\alpha \neq 0), \quad W_0(u) = \sum_g N_g \log u_g, \quad (3)$$

population-weighted by group size N_g ; $\alpha \rightarrow 1$ is utilitarian, $\alpha = 0$ is proportional fairness, and $\alpha \rightarrow -\infty$ approaches leximin. As detailed in Section V-A, this project identifies and corrects a measured pathology in the population-weighted form and adopts, by default, the egalitarian (population-unweighted) variant $W_\alpha(u) = \sum_g u_g^\alpha / \alpha$ instead, which counts each group once regardless of size.

IV. THEORETICAL RESULTS

Statements below preserve the status labels of the underlying project notes exactly: *proven* (theorem, proposition,

lemma, with a proof or complete proof sketch), *cited* (an external result relied upon), or *conjectural/conditional* (stated precisely, with the reason it is not yet proven). No conjectural or conditional statement is used as a premise for any theorem.

A. Submodularity of the recovery-only process

Lemma 1 (Layered live-edge representation). *Fix $\beta = 0$. Build a layered graph $\mathcal{L}$ on $V \times \{0, \dots, T\}$ with, for each round t , a self-edge $(v, t) \rightarrow (v, t+1)$ live iff $R_v^t = 0$, and a cross-edge $(u, t) \rightarrow (v, t+1)$ for each $(u, v) \in E$, live iff $X_{uv}^t = 1$. Then for every realization of the randomness, $A_t = \{v : (v, t) \text{ reachable from } S \times \{0\} \text{ along live edges}\}$.*

Proof. By induction on t ; at $t = 0$ both sides equal S , and the inductive step is exactly the layered-semantics transition rule read as a reachability statement. ■

Lemma 2 (Recovery alone preserves monotone submodularity). *For $\beta = 0$ and any recovery rates $q_v \in [0, 1]$, $f_0(S)$ is monotone and submodular in S .*

Proof. For each fixed realization, $|A_t|$ is the number of layer- t nodes reachable from $S \times \{0\}$ in a fixed DAG (Lemma 1) – a coverage function of S , hence monotone submodular; sums over t and expectations over realizations of monotone submodular functions remain monotone submodular. ■

This isolates exactly which relaxation is “free”: spontaneous, state-independent recovery is harmless (it is a random self-edge and reachability stays a coverage function), while neighbor-dependent deactivation (backfire) destroys the coverage structure, since an active in-neighbor can now *remove* a node from the reachable set. This sharpens the picture in Chan and Ning [20], who lose submodularity under non-progressive *threshold* rules where deactivation depends on neighbor states; recovery alone needs no such structure to stay submodular, on any graph.

B. A sandwich approximation for all backfire intensities

Let $U(S) := f_0(S)$ (the $\beta = 0$ process at the true recovery rates). For node v , let $\delta_v := 1 - \prod_{u \in N_{\text{in}}(v)} (1 - \beta p_{uv}) \leq \beta \Delta_{\text{in}}$, where $\Delta_{\text{in}} := \max_v \sum_{u \in N_{\text{in}}(v)} p_{uv}$, and let $L(S)$ be the $\beta = 0$ process under inflated recovery $q'_v := 1 - (1 - q_v)(1 - \delta_v)$.

Lemma 3 (Upper coupling). *Under shared randomness, $A_t^\beta \subseteq A_t^0$ for all t ; hence $f_\beta(S) \leq U(S)$.*

Lemma 4 (Lower coupling). *There is a coupling under which $A_t^\beta \subseteq A_t^0$ for all t ; hence $L(S) \leq f_\beta(S)$.*

Both couplings are proven by induction, replacing the per-edge backfire trials with a single uniform draw compared against the aggregate kill probability and, respectively, against δ_v ; the key inequality is that β -kill implies L -extra-recovery because the β -kill threshold is at most δ_v (attained when every in-neighbor is active).

Theorem 1 (Sandwich approximation, all β). *L and U are monotone submodular (Lemma 2) and $L \leq f_\beta \leq U$ pointwise (Lemmas 3–4). Let S_L, S_U be the greedy solutions for L, U under $|S| \leq k$, S_β the greedy solution run directly on f_β , and*

$S_{\text{sand}} = \arg \max_{S \in \{S_L, S_U, S_\beta\}} f_\beta(S)$. Then, with S^ optimal for f_β ,*

$$f_\beta(S_{\text{sand}}) \geq \max \left\{ \frac{f_\beta(S_U)}{U(S_U)}, \frac{L(S^*)}{f_\beta(S^*)} \right\} \left(1 - \frac{1}{e}\right) f_\beta(S^*) \geq \rho \left(1 - \frac{1}{e}\right) \text{OPT}_\beta \quad (4)$$

where $\rho := \min_{|S| \leq k} L(S)/U(S)$.

Proof. Cited: the sandwich-approximation theorem of Lu, Chen, and Lakshmanan [11], applied to the bracket $L \leq f_\beta \leq U$; the $(1 - 1/e)$ factors are [2] applied to greedy on each submodular surrogate. The final inequality follows from $L(S^*)/f_\beta(S^*) \geq L(S^*)/U(S^*) \geq \rho$. ■

Theorem 1 improves on treating backfire as an arbitrary perturbation from submodularity via [30], which would only hold for $\beta = O(n^{-1/2})$ – far below the $\beta \in [0.1, 0.7]$ regime our empirical study targets – because it exploits the *structure* of the perturbation (a bounded extra kill rate) rather than treating it adversarially, and it holds for every β , recovering the tight $(1 - 1/e)$ exactly at $\beta = 0$ (where $L = U$, $\rho = 1$). It also yields a practical corollary: since L and U are ordinary $\beta = 0$ processes that can be simulated directly, $f_\beta(S)/U(S)$ is a directly computable, per-instance certificate, $f_\beta(S) \geq \frac{f_\beta(S)}{U(S)} (1 - \frac{1}{e}) \text{OPT}_\beta$ – the algorithm ships its own approximation guarantee on every instance it is run on.

Proposition 1 (Finite-horizon bound on ρ). $\rho \geq (1 - \beta \Delta_{\text{in}})^T$.

This is proven by charging every self-edge of a witness reachability path under the coupling of Lemma 4, and is conservative (it ignores re-activation through alternate paths). A tighter steady-state conjecture, $\rho \gtrsim q/(q + \beta \Delta_{\text{in}})$ for $q = \min_v q_v$, is stated in the underlying project notes as an explicit *conjecture*: the heuristic single-sojourn argument does not account for the downstream cascade a shortened sojourn fails to seed, and closing that gap is left open. This conjecture is not used in any proof in this paper (Section VIII discusses why it is, in any case, not the most promising target for tightening the sandwich ratio).

C. A closed-form index for the control problem

Given the mean-field belief field (Section V), each node’s per-round control problem is a 2-state MDP – states $s \in \{0, 1\}$, discount γ , reward w at $s = 1$ – with natural transitions $a = p_{01}$ (activate), $b = p_{10}$ (deactivate), and three actions: NONE (free, either state), CONVERT (cost c_C , forces $0 \rightarrow 1$, only at $s = 0$), MAINTAIN (cost c_M , forces stay-at-1, only at $s = 1$), under a shared Lagrange price λ per unit cost. There are exactly four deterministic stationary policies, named by (action at 0, action at 1): NN, NM, CN, CM.

Proposition 2 (Closed forms). *With $D := (1 - \gamma)(1 - \gamma + \gamma(a + b))$, each policy’s value functions $V_1(\lambda), V_0(\lambda)$ are given in closed, affine-in- λ form by solving its fixed 2×2 linear system (e.g., for CN, $V_0 = -\lambda c_C + \gamma V_1$ and $V_1 = w + \gamma[(1 - b)V_1 + bV_0]$, giving $V_1 = (w - \gamma b \lambda c_C)/((1 - \gamma)(1 + \gamma b))$); the full table of four closed forms is given in the underlying project notes and reproduced in the algorithm’s implementation.*

Proposition 3 (Convexity in λ). $V^*(s, \lambda) = \max_\pi V_\pi(s, \lambda)$ over the four policies is a maximum of four affine functions

of λ : convex, piecewise linear, with at most three breakpoints, computable in $O(1)$.

Proof. In a discounted finite MDP some stationary deterministic policy is optimal at every state simultaneously, so V^* is exactly this finite max; Killian, Perrault, and Tambe [41] (their Prop. 4.1) prove convexity in λ for the general multi-action restless-bandit case, of which this is a one-line special case via finite enumeration of the four policies. ■

Proposition 4 (Index and indexability, this case). *Define node v 's index λ_v^* as the largest $\lambda \geq 0$ at which the optimal policy still takes the paid action at v 's current state.*

A proof *sketch* shows the paid action's advantage over NONE has a strictly negative slope in λ from the cost term, while the value gap $V_1^* - V_0^*$ is non-increasing in λ on each linear piece, checked by finite case analysis over the four policies. **We state plainly, matching the underlying project notes' own status label, that indexability – i.e., that the paid set is a down-set $[0, \lambda_v^*]$ – is only verified numerically here** (on random instances, over a fine λ grid, checking that per-node cost is non-increasing in λ), not proven in general; the finite four-policy structure makes this tractable to check exactly where Killian et al. [41] call general multi-action indexability "notoriously difficult" and avoid it. The index itself is defined as the supremum of the paid set, which remains well-defined and exactly computable regardless.

Theorem 2 (Per-round complexity). *Given Propositions 2–4, one round of the control policy costs $O(Km + n \log n)$, where K is the number of mean-field fixed-point iterations (geometric convergence under a contraction condition on $\Delta \max(p^+, p^-)$).*

The $n \log n$ term sorts nodes by λ_v^* and fills the shared budget greedily – exactly "smallest λ with aggregate cost $\leq B$," the target of the bisection-plus-value-iteration procedure it replaces, which cost $O(n \cdot I_{\text{bisect}} \cdot I_{\text{VI}})$ (on the order of 8000 n scalar operations per round in our configuration); this project's own measurements put the closed-form replacement at roughly $130\times$ faster per round than that bisection oracle. For comparison, naive Monte Carlo KKT greedy costs $O(knRm)$ per seed set, and one static IMM selection costs $O((k+\ell)(n+m) \log n/\varepsilon^2)$ [5]; our per-round cost is near-linear like a single IMM run but is paid T times, the stated price of sustaining influence under decay rather than seeding once.

D. Hardness

For $\beta = 0$ (and $q = 0$, large T), the problem contains max- k -cover, so $1 - 1/e$ is an unbeatable ceiling for every $\beta \geq 0$ under $P \neq NP$ [3], [2] – adding backfire cannot remove hard instances on which β is simply irrelevant. For adversarial, arbitrary non-submodular perturbations, no constant-factor approximation exists in general once a function is $\omega(n^{-1/2})$ -far from submodular [30]; Theorem 1 escapes this only because our perturbation is structured, not adversarial.

Theorem 3. *For every fixed $\beta \in [0, 1)$, no polynomial-time algorithm approximates f_β within $(1 - 1/e + \varepsilon)$ for any $\varepsilon > 0$ unless $P = NP$.*

Proof. Reduction from max- k -cover, $(1 - 1/e + \varepsilon)$ -inapproximable unless $P = NP$ [3]. Build a bipartite digraph (set-nodes to element-nodes, edge probability 1), each element replicated to swamp the additive seed-count term. The key subtlety is that backfire *is* realized on this instance (a seed keeps attacking every element it covers), so its harmlessness must be shown, not assumed. On this instance class, an element with $c \geq 1$ active covering seeds throughout is a two-state chain with up-rate 1 and down-rate $1 - (1 - \beta)^c$, giving an exact value formula $f_\beta(S) = kT + T \sum_v \varphi_\beta(c_v(S)) + O(n)$ with $\varphi_\beta(c) = 1/(2 - (1 - \beta)^c)$ strictly decreasing in $c \geq 1$ and bounded below by $1/2$; the max-coverage gap therefore survives backfire intact. ■

This value formula also shows $f_\beta \geq f_0/2$ for every seed set on this instance class at $q = 0$, exactly where the sandwich ratio ρ of Theorem 1 is vacuous ($\rho = 0$).

A natural next question is whether backfire forces a further, β -dependent worsening beyond $1 - 1/e$. We do not claim such a result here: a candidate constant $c(\beta) \in [0.94, 1)$ is derivable under one extra, unverified property of Feige's hard instances, but proving that property would require re-deriving part of Feige's reduction, which we have not done to a standard we are willing to publish as a theorem. We report this only as an open conjecture, with supporting (not conclusive) structural and numerical evidence, in Section VIII.

V. ALGORITHM: MF-BWI-FAIR

MF-BWI-Fair (Mean-Field Belief with a Lagrangian index and isoelastic-Welfare reweighting, Fair) is a stateful per-round control policy. Each round:

- 1) **Bayesian posterior tracking.** A Beta-Bernoulli tracker maintains posterior means $\hat{p}_{uv}^+, \hat{p}_{uv}^-, \hat{q}_v$ for every edge/node from observed trial outcomes; the policy never sees the simulator's hidden true parameters, only these running estimates.
- 2) **Mean-field belief fixed point.** Exact per-node marginals require tracking correlations across the whole active-set distribution, intractable at scale. We use the standard individual-based / N-intertwined mean-field approximation from network epidemic models: each node's in-neighbors are replaced by their marginal activation probabilities m_u , giving the self-consistency equation

$$m_v = m_v(1 - \hat{q}_v) \prod_{u \in N_{\text{in}}(v)} (1 - \hat{p}_{uv}^- m_u) + (1 - m_v) \left(1 - \prod_{u \in N_{\text{in}}(v)} (1 - \hat{p}_{uv}^+ m_u) \right), \quad (5)$$

solved by Jacobi fixed-point iteration to tolerance 10^{-4} or 50 iterations. We verified directly that the alternative network-coupling method of Ou et al. [42] does not apply here, since it requires a deterministic action-vector coupling rather than the genuinely bilinear neighbor-state coupling this model has.

- 3) **Group welfare weights.** Each group's current marginal welfare weight w_g is computed from its tracked time-averaged realized reach u_g (Section V-A), egalitarian by default.

Algorithm 1 MF-BWI-Fair, one round

Require: graph G , state s , budget B , fairness exponent α

- 1: $(\hat{p}^+, \hat{p}^-, \hat{q}) \leftarrow$ posterior means from tracker
- 2: $m \leftarrow$ mean-field fixed point of Eq. (5) from $(s, \hat{p}^+, \hat{p}^-, \hat{q})$
- 3: $(p_{01}, p_{10}) \leftarrow$ field transition probabilities from m
- 4: $u \leftarrow$ tracked per-group time-averaged reach
- 5: $w \leftarrow \text{GROUPWELFAREWEIGHTS}(u, \alpha)$ (egalitarian default)
- 6: **for** each node v **do**
- 7: build 2-state/3-action MDP with reward $w_{g(v)} \cdot s_v$, kernel $(p_{01}(v), p_{10}(v))$
- 8: $\lambda_v^* \leftarrow$ closed-form index (Prop. 4)
- 9: **end for**
- 10: sort nodes by λ_v^* descending; select prefix fitting in B
- 11: greedily fill leftover budget by descending index among unselected nodes
- 12: apply CONVERT/MAINTAIN to selected nodes, NONE elsewhere
- 13: observe transition, update Bayesian tracker and u_g

- 4) **Closed-form Lagrangian index.** Using the converged belief field to obtain each node's "no-action" transition probabilities $p_{01}(v), p_{10}(v)$, each node's 2-state/3-action MDP is built with reward $r_v(s) = w_{g(v)} \cdot s$, and the shared-budget Lagrangian relaxation is solved via the closed-form index of Section IV-C (a bisection-plus-value-iteration oracle is kept as a test reference).
- 5) **Leftover-budget fill.** Any budget left over from the index scan's discreteness is filled by a greedy tie-break on each node's closed-form index (or Bellman gap, under the bisection oracle), keeping the hard budget constraint intact without reintroducing per-group floors or quotas.
- 6) **Execute, observe, update.** Actions are applied; the simulator advances one round under the layered-semantics transition rule (Section III); observed trial outcomes update the Bayesian tracker; and each group's realized reach average u_g is updated for the next round's weights.

A. The fairness fix: population-weighted vs. egalitarian welfare

Rahmattalabi et al.'s isoelastic welfare weight, implemented exactly as published, is $w_g = N_g u_g^{\alpha-1}$ (or N_g/u_g at $\alpha = 0$): the correct marginal utility for maximizing *aggregate* welfare, in which a group's contribution scales with its population. This has a concrete, measured downside for the stated goal of not neglecting a small group: at $\alpha = 0$, group g 's weight exceeds group g' 's only once $u_{g'}/u_g > N_{g'}/N_g$ – the smaller group must be under-served by a factor proportional to the group-size *ratio* before the mechanism reacts at all. On this project's own three-group (20/40/60) benchmark graph, at a neutral $\alpha_{\text{fair}} = 0$ this kept the smallest group at a visibly lower realized reach than the largest purely because a $3\times$ reach gap was required before the weight formula began favoring it. We therefore adopt, as the default, the egalitarian (population-unweighted) form $w_g = u_g^{\alpha-1}$ (or $1/u_g$ at $\alpha = 0$), a standard

alternative welfare form in the social-welfare and fair-division literature (not a new mechanism), counting each group once regardless of size – the appropriate choice when fairness means each *group*, not each person, is treated fairly regardless of population.

VI. EXPERIMENTAL SETUP

Graph. The primary validation graph is a single synthetic stochastic block model with three groups of unequal size (20/40/60, 120 nodes total), within-block edge probability $p_{\text{in}} = 0.07$, between-block edge probability $p_{\text{out}} = 0.008$, and per-edge positive-influence probability $p_{uv} \sim \text{Uniform}(0.05, 0.15)$. Density was hand-tuned so the network neither saturates near full activation nor fails to cascade at all, either of which would mask the effects under study. This is the primary benchmark reported in Section VII-A; Section VII-B additionally reports the same six-algorithm comparison on three further synthetic graphs at different scales and topologies and on one real social network, to check that the main benchmark's qualitative finding is not an artifact of this one hand-tuned graph (experiments/MULTIGRAPH_VALIDATION.md).

Budget and horizon. Per-round budget $B = 100$, horizon $T = 30$ rounds. One-shot baselines are given $k = 20$ seeds, chosen so that $k \cdot \text{cost}(\text{CONVERT}) = 100 = B$: one round's worth of MF-BWI-Fair's budget, spent once at round 0.

Baselines. *KKT-greedy/CELF* [1], [4]: classical Monte Carlo greedy under plain progressive IC ($\beta = 0, q = 0$), this project's own reimplement, used as the "ignore all four relaxations" ablation. *Fair-greedy*: an in-house welfare-weighted greedy baseline in the style of [10] maximizing a sum of concave per-group log-utilities. *Repeated-greedy*: an in-house cost-effective greedy that re-selects actions every round under the same per-round budget, using *true* parameters directly (no Bayesian uncertainty, no fairness mechanism) – the strongest plausible classical competitor that also acts repeatedly, included specifically to isolate whether MF-BWI-Fair's advantage is more than "gets to act every round." *IMM* [5]: this project's independent reimplement of the two-phase martingale sampling and node-selection algorithm, cross-checked against the paper's own pseudocode and against a documented erratum [12]. *He–Kempe Saturate Greedy* [31]: this project's independent reimplement of the bicriteria robust algorithm, given the low/high corners of the p_{uv} range as its scenario set. *MF-BWI-Fair*, as described in Section V.

Honesty notes on baseline provenance. Other than the objective forms they target, every baseline in this comparison is an independently-verified reimplement, not the original authors' code. Official code exists for IMM (a SourceForge project by the original authors) and no official code was found for He–Kempe's Saturate Greedy or for Rahmattalabi et al.'s welfare mechanism specifically. All IMM numbers reported here are from this project's own Python reimplement; we additionally benchmark that reimplement directly against the original authors' C++ implementation and report the outcome in Section VII-C (full method in experiments/IMM_VALIDATION.md). No equivalent direct comparison exists for He–Kempe or Rahmattalabi et al.'s

TABLE I
TIME-AVERAGED SPREAD, 120-NODE SBM BENCHMARK (15 TRIALS).
SG = HE-KEMPE'S SATURATE GREEDY.

Algorithm	$\beta=0, q=0$	$\beta=0.7, q=0.05$	$\beta=0.15, q=0.4$
KKT-greedy/CELF	75.7	51.9	12.9
Fair-greedy	76.9	51.9	12.9
IMM	78.0	52.5	15.4
He-Kempe SG	76.1	47.6	13.4
Repeated-greedy	105.0	93.9	74.4
MF-BWI-Fair	104.2	101.3	96.8

mechanisms, since no official code was found for either; those numbers remain characterizations of this project's own reimplementations only, cross-checked against the papers' stated algorithms and theorem statements, not the original authors' published numbers.

Metrics. (a) time-averaged active-node count over the full horizon; (b) per-group reach fraction (own-group-size-normalized, time-averaged) and its minimum across groups; (c) wall-clock runtime. Sweeps are conducted over backfire intensity β , recovery rate q , and the fairness exponent α_{fair} , each with 15 independent trials using common random numbers across the swept parameter within a trial to reduce sweep-trend noise without narrowing between-trial variance.

VII. RESULTS

A. Main benchmark: a 120-node stochastic block model

Table I reports time-averaged spread for all six algorithms at the extremes of the backfire (β) and recovery (q) sweeps described in Section VI; the full sweeps (six β points, five q points, 15 trials each) show the same pattern monotonically in between. At $\beta = q = 0$ – the plain progressive-IC setting every one-shot baseline targets, and where neither MF-BWI-Fair nor repeated-greedy can exceed the proven $(1-1/e)$ ceiling – MF-BWI-Fair and repeated-greedy are statistically tied (within 1% across repeated runs under independently re-derived random seeds); the moment either β or q departs from zero, MF-BWI-Fair opens a lead over every other algorithm that grows monotonically with the relaxation's severity, reaching more than $6\times$ the best one-shot baseline's spread at $q = 0.4$.

The four one-shot classical baselines (KKT-greedy, Fair-greedy, IMM, He-Kempe robust) cluster together throughout, as expected: none of them is designed to act more than once or to track a changing state, so all four degrade together as backfire/recovery intensify, regardless of which classical guarantee they individually carry. He-Kempe's Saturate Greedy is consistently the weakest of the four (it hedges against a worst-case scenario corner that never actually realizes in this benchmark, spending roughly 25% fewer seeds – about 15 of the $K = 20$ budget – to do so); this is the correct, expected behavior of a robustness guarantee under a benign instance, not a defect. MF-BWI-Fair's margin over He-Kempe specifically is the cleanest evidence that *paying for robustness against a probability range and adaptively learning the realized value inside that range* are not the same thing: MF-BWI-Fair beats He-Kempe by a wide margin even at $\beta = q = 0$ (104.2 vs. 76.1), where it is merely tied with repeated-greedy.

Fairness (min-group reach, the minimum own-group-normalized reach across groups) tells a related story once the population-weighting pathology in the isoelastic welfare objective is corrected (Section V): at the neutral default, MF-BWI-Fair matches or leads repeated-greedy's min-group reach almost everywhere ($\beta = 0$: 0.856 vs. 0.849; $\beta = 0.7$: 0.824 vs. 0.766; $q = 0.4$: 0.809 vs. 0.554), with a single small residual gap at $q = 0$ (0.860 vs. 0.881). Total spread is essentially unchanged by the fairness correction, so this is a genuine improvement in the default operating point rather than a trade-off paid for it. The α_{fair} knob provides an additional, independent lever on top of this default: min-group reach rises monotonically from 0.777 at the utilitarian extreme ($\alpha_{\text{fair}} = 1$) to 0.856 at a leximin-like extreme ($\alpha_{\text{fair}} = -8$), crossing repeated-greedy's fixed 0.803 by $\alpha_{\text{fair}} \approx 0.5$, with total spread *rising*, not falling, across the same range – fairness and spread are not in tension in this benchmark. On the runtime axis, the closed-form Lagrangian index (Section V) is approximately $130\times$ faster per per-round decision than the bisection-plus-value-iteration procedure it replaces, at identical output (verified on matched inputs), making MF-BWI-Fair's per-round cost $O(Km + n \log n)$ rather than the multiplicatively more expensive bisection oracle.

B. Multi-graph and real-network validation

The benchmark above uses a single hand-tuned synthetic graph. We validate generalization on three further synthetic graphs – `small_sbm` (60 nodes), `large_sbm` (240 nodes, double every block size of the main benchmark), and `barabasi_albert` (150 nodes, a structurally different scale-free topology) – and on one real social network, `real_facebook_348`, a 224-node, 6,384-edge connected friendship subgraph from the SNAP ego-Facebook dataset [46] (Weighted-Cascade edge probabilities [47], and community-detection-derived groups as a structural fairness proxy in the absence of demographic attributes). Full detail, including the reduced compute budget used for this validation pass and every raw per-trial result, is reported in `experiments/MULTIGRAPH_VALIDATION.md`.

The synthetic-graph finding reproduces across scale and topology. On all three additional synthetic graphs, MF-BWI-Fair is tied with repeated-greedy at $\beta = q = 0$ and opens a clear, growing lead as β or q increases, in both spread and min-group reach – e.g. `large_sbm` at $q = 0.4$: 202.8 vs. 164.2 spread (+23%); `small_sbm` at $q = 0.4$: min-group reach 0.772 vs. 0.553. This result is not an artifact of the one hand-tuned 120-node graph used for the main benchmark.

The real network initially diverged, and restoring compute budget resolved it. At a necessarily reduced compute budget (shorter horizon, smaller K/B , fewer Monte Carlo samples, and critically a cut-down 1-rollout repeated-greedy instead of its 2-rollout library default – required to keep runtime tractable at this graph's ~ 28.5 average degree, 4–9 $\times$ every synthetic graph tested), MF-BWI-Fair did *not* show the dominant advantage seen on every synthetic graph: it trailed repeated-greedy by 2–11% in spread at every checkpoint tested and showed a mixed, sometimes clearly

TABLE II
REAL FACEBOOK EGO-NETWORK (REAL_FACEBOOK_348):
MF-BWI-FAIR VS. REPEATED-GREEDY, REDUCED VS. FULL COMPUTE
BUDGET.

Checkpoint	Reduced budget	Full budget
$\beta = 0.0$, spread	−9.3%	−1.2% (tied)
$\beta = 0.5$, spread	−5.1%	+1.5%
$\beta = 0.5$, min-reach	−26%	+3.8%
$q = 0.1$, spread	−4.2%	−0.2% (tied)
$q = 0.4$, spread	−2.3%	+2.5%
$q = 0.4$, min-reach	+14%	+4.3%

worse, min-group reach (e.g. $\beta = 0.5$: 0.351 vs. 0.474, a 26% deficit). We report this plainly, as an honest negative finding, rather than omitting it. To disentangle whether this was a genuine structural effect of the graph’s density on MF-BWI-Fair’s mean-field approximation, or simply an artifact of the reduced compute budget applied to make the validation tractable, we re-ran the identical graph with the compute budget fully restored to the main benchmark’s settings ($T = 30$, $B = 100$, $K = 20$, 15 trials, full Monte Carlo sample counts, repeated-greedy’s own 2-rollout default) – see `experiments/REAL_GRAPH_FULLBUDGET_FOLLOWUP.md`. Every one of the four original checkpoints flips from “MF-BWI-Fair behind” to “tied or ahead” once the budget is restored (Table II), and the qualitative pattern – tied at low backfire/recovery, a growing MF-BWI-Fair lead as either increases – matches the synthetic-graph family. The margin on this real, much denser graph is smaller than on any synthetic graph tested (e.g. +2.5% spread at $q = 0.4$, vs. +23% on `large_sbm` at the same q), consistent with a modest, non-reversing cost of the mean-field decoupling approximation on a high-average-degree real network – a limitation worth flagging (Section VIII), not one that overturns the core finding.

C. Validation against the original authors’ IMM implementation

Every baseline reported above other than the classical objective forms they target is this project’s own reimplementations, not the original authors’ code (Section VI). We benchmark our IMM reimplementations [5] directly against the original authors’ published C++ implementation (full method in `experiments/IMM_VALIDATION.md`) on the main benchmark graph. The two implementations select different-but-comparable seed sets (low node-level overlap is expected on a small, community-structured graph with substantial within-block redundancy), but land within 0.6% of each other’s independently re-measured expected spread (32.08 vs. 32.27) and within 2% of CELF greedy’s spread – consistent with both legitimately approximating the same submodular objective well, as IMM’s $(1 - 1/e - \varepsilon)$ -approximation guarantee promises. No bug was found in our IMM reimplementations during this exercise. As expected from IMM’s near-linear-time design, the original C++ binary is faster still than our Python port at this graph’s small scale (sub-millisecond vs. 0.019s), both far faster than CELF’s Monte Carlo greedy (1.8s); IMM’s

design advantage is intended to show up at much larger scale than this benchmark’s 120 nodes.

VIII. DISCUSSION AND LIMITATIONS

Graph coverage remains limited, and the one real network tested shows a narrower margin. Section VII-B shows the main benchmark’s finding reproduces cleanly across two further synthetic scales and a structurally different scale-free topology, and – once compute budget is matched to the main benchmark – on one real social network, in the same qualitative shape (tied at low backfire/recovery, growing MF-BWI-Fair lead as either increases). But this is still a single real network ($N=1$ at the level of “real networks tested”), and its margin is smaller than any synthetic graph tested (e.g. +2.5% spread at $q = 0.4$ vs. +23% on `large_sbm`), a gap in magnitude, though not direction, that is a plausible early signature of the mean-field decoupling approximation (Section V) degrading somewhat on real, much denser, more clustered networks – consistent with, but not confirmed by, the next limitation below. Broader real-network coverage and degree-distribution families beyond this one ego-network remain future work.

A further backfire-dependent hardness worsening remains an open conjecture, deliberately not claimed as a result. A candidate constant $c(\beta) \geq 0.94$ (Section IV-D) would follow from one extra property of Feige’s hard instances – that the fraction of elements covered exactly once by any candidate solution is bounded by roughly $1/e$ – which is plausible (a concentration argument over Feige’s random-partition construction) but requires two technical steps we have not both closed: (i) a concavity property of $d \mapsto \mathbb{E} \varphi_\beta(\text{Bin}(d, 1/k))$, where $\varphi_\beta(c) = 1/(2 - (1 - \beta)^c)$ is the exact per-element value function from Theorem 3’s proof; (ii) a cross-term pairwise-soundness property of the same reduction, re-verified for this statistic rather than for plain coverage. We checked (i) numerically – exhaustively across $\beta \in [0.01, 0.99]$ and $k \in \{2, \dots, 200\}$, it holds for every degree $d \leq k$ (the regime Feige’s construction uses) and only fails for $d \gtrsim 4k$, well outside that regime – which is genuine evidence, not a proof, since it does not establish the claim analytically and (ii) is a separate, deeper re-derivation of part of Feige’s reduction that we have not attempted to a publishable standard. We therefore report this only as a conjecture with supporting structural evidence (the per-edge revive-at-least-as-fast-as-kill asymmetry noted in Section IV-D, and several natural gadget constructions all failing to encode a hard-to-decide dependence on the seed set) and the numerical check above, not as a theorem or a proposition of this paper.

No comparison against the original authors’ code for He-Kempe or Rahmattalabi et al. Neither paper has published code corresponding to the specific algorithm reimplemented here (a superficially related repository for Rahmattalabi et al. in fact implements a different paper, Tsang et al. [13], and must not be conflated with it). Only IMM has verifiable original code; Section VII-C reports a direct benchmark against it and finds our reimplementations trustworthy, but this does not extend to He-Kempe or Rahmattalabi et al., whose

reported numbers here remain characterizations of independent reimplementations only.

The mean-field approximation’s own error is not separately quantified. The belief fixed point of Section V is an approximation whose deviation from the true (intractable) joint activation distribution has not been bounded or measured in isolation from the end-to-end policy’s performance; this is a natural target for future analysis.

The sandwich bound is demonstrably loose, and this is a direction, not a hidden flaw. Exact computation on small instances (2–8 nodes, $T \sim 300$) shows the sandwich ratio ρ loose by up to an order of magnitude where it applies at all (e.g., roughly $13\times$ on a directed star instance, because the lower surrogate inflates every node’s recovery uniformly by δ_v even where a node has no active in-neighbor left), and vacuous whenever $q = 0$ or $q + \beta\Delta_{\text{in}} \geq 1$. Separately, f_β/f_0 is not bounded below by any instance-independent constant (a bidirected K_6 instance at $\beta = 0.5, q = 0.05$ collapses by a factor of roughly $300\times$ relative to f_0), so no purely multiplicative comparison against f_0 can be the source of a strong general guarantee. Yet greedy run on the plain $\beta = 0$ surrogate landed within 1% of OPT_β on every one of the six small instances tested – suggestive, not proof, that the practical quality of surrogate-greedy substantially exceeds what $\rho(1-1/e)$ certifies. We read this as the concrete next step for this line of work: a sharper *upper*-bound analysis (a per-instance or instance-class argument in the spirit of the exact layered value formula behind Theorem 3), rather than a search for a matching lower bound on ρ itself, which the evidence here suggests is not the right quantity to target.

Indexability is checked, not proven. As stated in Proposition 4, the down-set property underlying the closed-form index’s exactness is verified numerically on random instances over a fine λ grid, not established as a general theorem for this 2-state/3-action case.

An open, unconditional route to a matching lower bound. We identify, but do not pursue to completion, a possible unconditional alternative to the conjectured constant above: the class of functions $S \mapsto \sum_v \varphi(c_v(S))$ is closed under the same symmetrization used in Vondrák’s symmetry-gap framework [48], and the symmetric-instance gap for φ_β (Theorem 3’s proof) works out numerically close to, and slightly below, our conjectured $c(\beta)$. Whether Vondrák’s smoothing-step argument – which uses submodularity directly – extends to this non-submodular but structured class is open and was not checked; if it does, it would give an unconditional oracle-model lower bound without needing the unverified property the conjecture above depends on. We flag this as the single most promising next step toward a proof, should this line of work be pursued further.

IX. CONCLUSION

We formalized a combined relaxation of the four assumptions behind the classical influence-maximization guarantee – progressive dynamics, monotone submodularity, known parameters, and fairness-free objectives – and proved that a sandwich approximation recovers a $\rho(1-1/e)$ guarantee for

every backfire intensity, with a closed-form index making the resulting control problem tractable at near-linear per-round cost. We proved that $(1-1/e)$ -hardness survives backfire exactly, while explicitly declining to upgrade a plausible but unproven constant-factor worsening result to theorem status. We corrected a measured fairness pathology in an existing welfare objective and implemented the result in MF-BWI-Fair. Experiments on the main benchmark show MF-BWI-Fair matching or exceeding every tested baseline, including independently-verified published algorithms, in both spread and fairness, and this finding reproduces across three further synthetic graphs spanning two scales and a structurally different topology. On the one real social network tested, the finding initially did not hold under a necessarily reduced compute budget, but restoring that budget on the identical graph flipped every tested checkpoint from behind to tied-or-ahead – evidence that the earlier divergence was substantially a budget artifact, though the resulting margin remains narrower on this real, denser network than on any synthetic graph tested, an open question for future work along with broader real-network coverage. We also directly validated our IMM reimplementation against the original authors’ C++ code and found it trustworthy.

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